# Conflict, Information, and the Logic of Power

Anton Sobolev

University of Augusta | Spring ’26

# Information in the Logic of Power

## Research Agenda & Historical Context

conflict

- **What does explain power dynamics?**

☐ Geography ☐ Institutions ☐ World Economy ☐ Prod. Technology

Diamond 1997, Ross 2001, Sachs 2003

North 1995, Acemoglu et al 2001

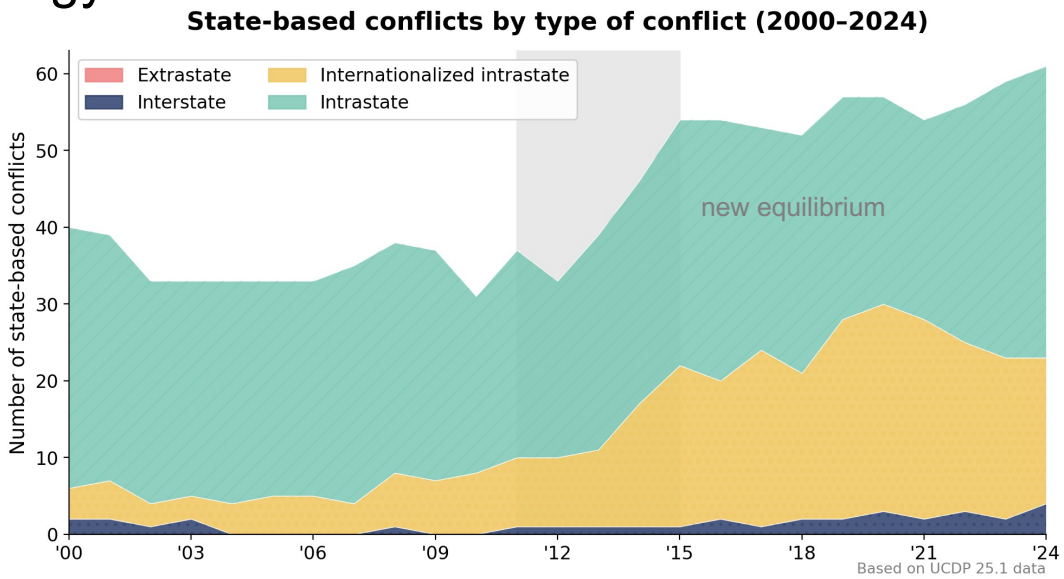
Rogowski 1989 Hiscox 2002

Rueschemeyer et al 1992, Boix 2003, Ben & Samuels 2014

- **Yet we observe** Starting 2010x

☐ Not a spike, but a transition to a new equilibrium

☐ Other things being equal?



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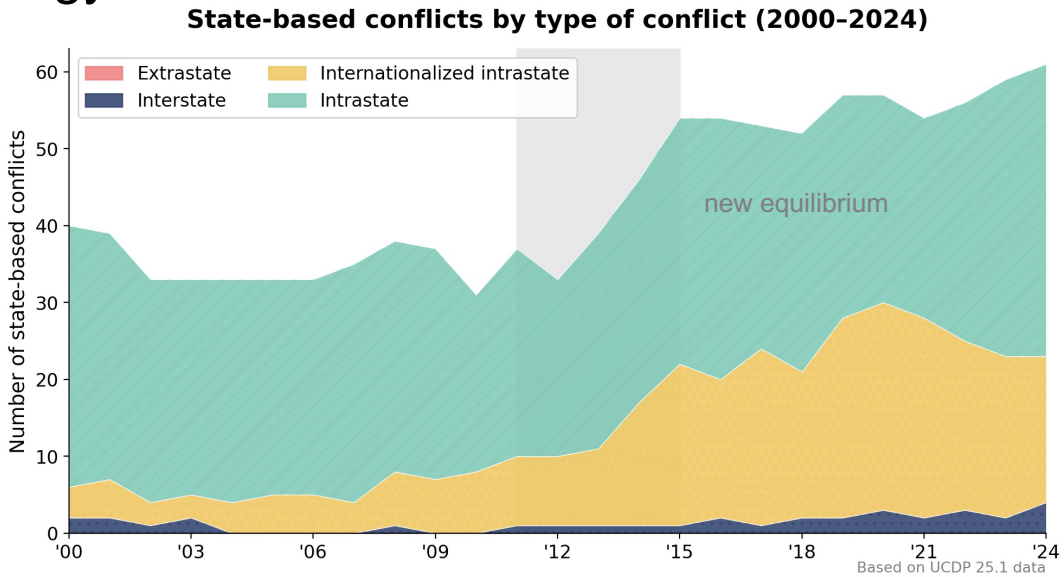


### ■ Yet we observe Starting 2010x

☐ Not a spike, but a transition to a new equilibrium

☐ Other things being equal?

### ■ Emerging Field Digital Tech in Politics

☐ Independent Role of ICT [≠ Propaganda] Pierskalla & Hollenbach 2013, Dagaev, Lamberova, Sobolev 2019, Nguyen 2024, Kuznetsova & Tolbert 2024

☐ Re-invention of Propaganda and Censorship King, Pan, Roberts 2013, Lorentzen 2014, Bhattacharya et al. 2023

☐ New types of Political ☐ Actors: Platforms, Pirate Parties, Influencers

Policy ☐ Agents: Bots (and Viruses), Trolls, Scams, APT groups

# Information in the Logic of Power

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- **Starting Point**
  - Geography

    Diamond (1997)
    Ross (2001)
    Sachs (2003)

    Campante & Do (2009),
    Campante (2012),
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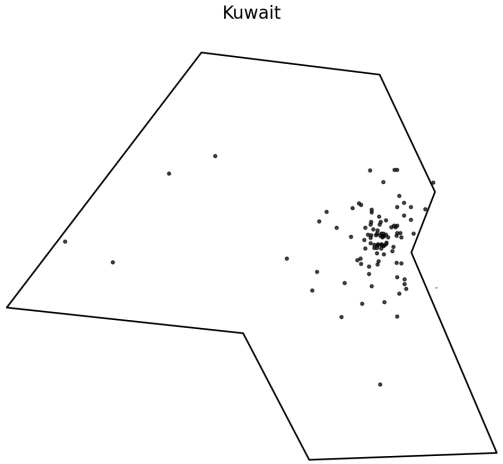
- **Case of Mass Actions** **2011-12, Russia, Regions**



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## ■ Starting Point

☐ Geography

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## ■ Case of Mass Actions 2011-12, Russia, Regions, Sobolev (2014)

☐ Expectations: ☐ Conflict is determined by geography and its derrivaties

☐ Population Allocation ☐ Climate ☐ Rough Terrain

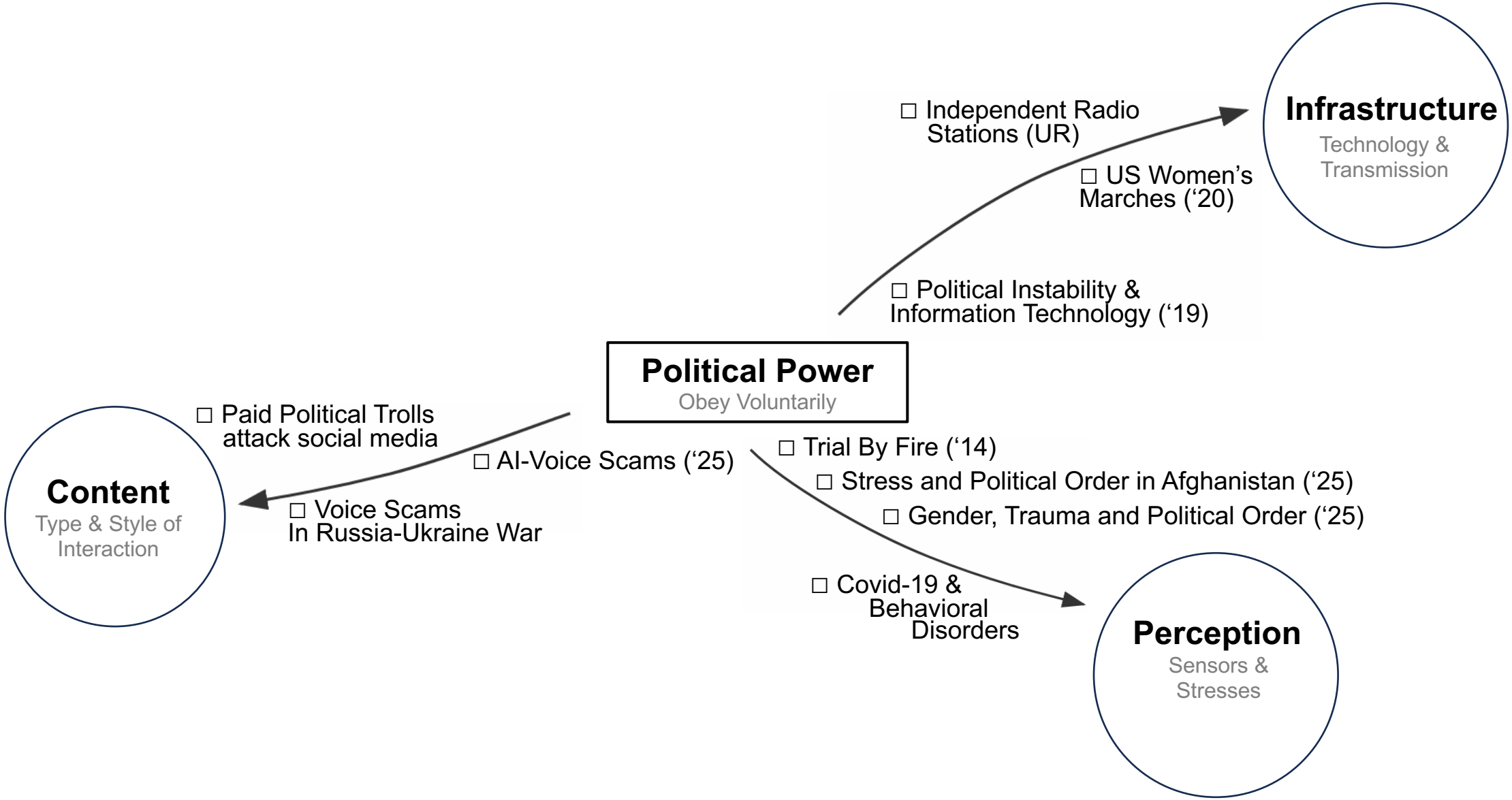
☐ Compare with: ☐ Economic Growth

☐ Unemployment ☐ *per capita* Growth

☐ Findings: ☑ Comparable Effects with Economic Factors

☑ ICT indicators (networks, computers, cellphones, information accessibility, internet traffic) **mattered even more!**

# Power & Information



# Power & Information

**Sidenote:** Where is Political Power?
A driver follows traffic rules [to avoid a ticket]

Follow the rules even if …

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**Second Law of Thermodynamics** (Clausius, Kelvin, Thomson)
Entropy irreversibly increases in isolated systems

**Biological Life** (Schrodinger 1944, Bertalanffy 1968)
Localized resistance to entropy through temporary ordering of matter

# Power & Information

**Legitimacy:** Alignment of agents with system rules without coercion

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Legitimacy locally but radically reduces entropy

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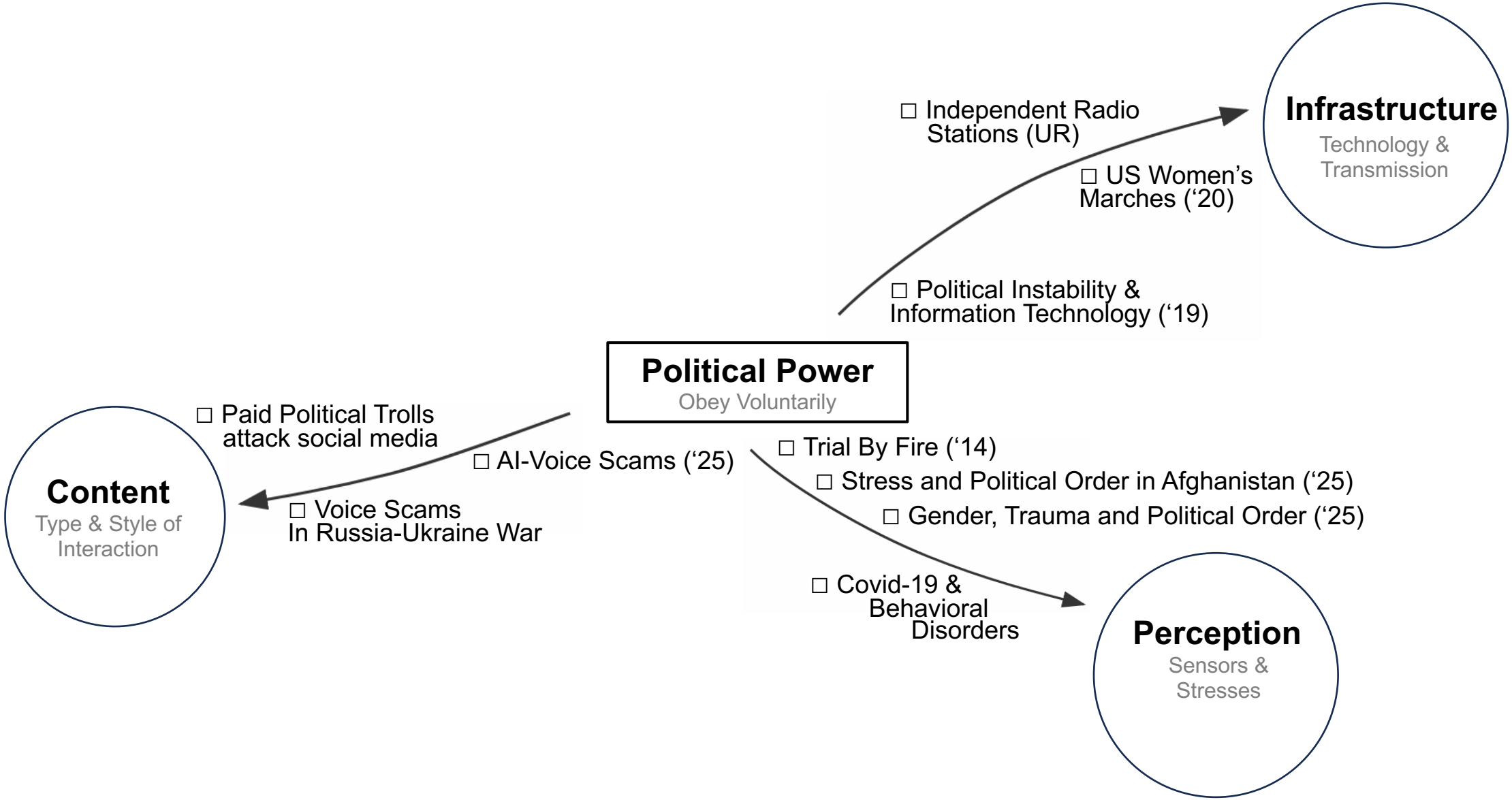
**No Legitimacy** ⇝ ~~**Political Power**~~ ⇝ **Coercion**

# Power & Information



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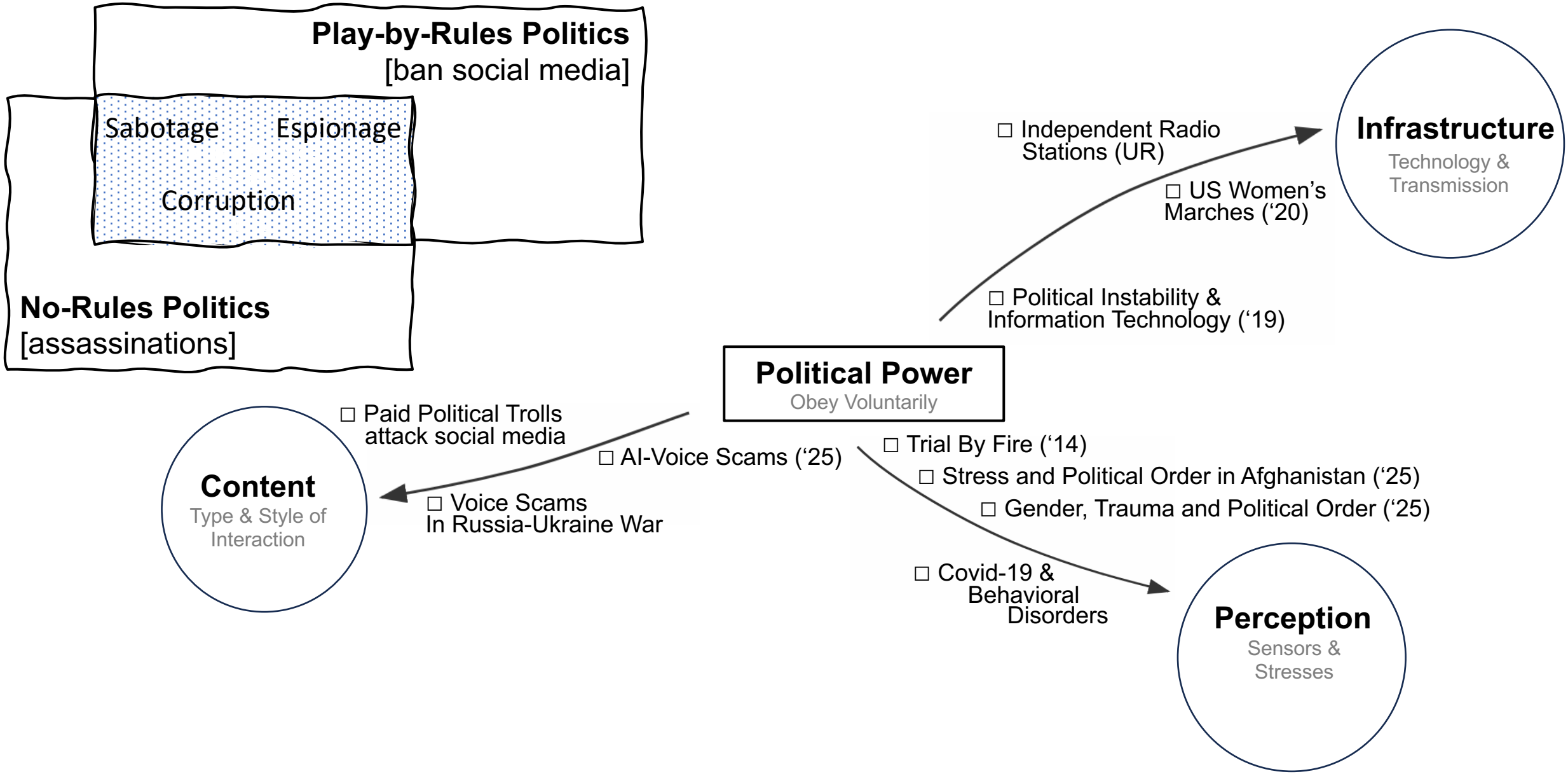

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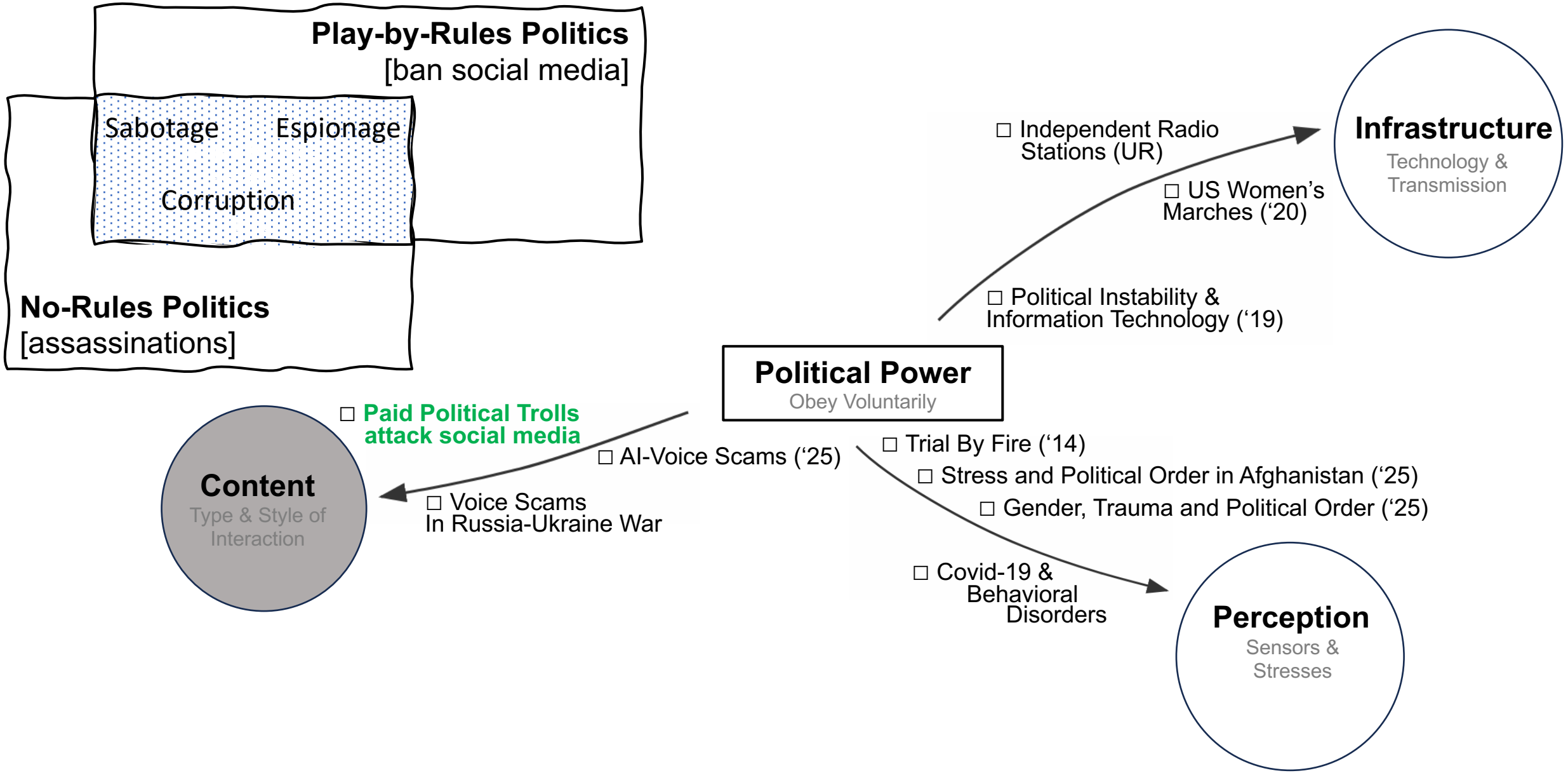

# Paid Political Commentators

## ☐ Setting

- **Motivation**
- **Definition:** Troll VS Paid Troll | Troll VS Bot
- **Context:** Authoritarian & Foreign Policy

## ☐ Challenges

- **Flexibility:** Strategies & Goals
- **Accounts Identification**
- **Theory**: Goals ≠ Incentives ≠ Effects
- **Causality:** Strategic Selection of posts / conversations

# 1 Trolls as a Cyber Weapon

## Flexibility & Customization

- **Levels**
  - **Macro:** Manipulate platform algorithms
  - **Micro:** Threaten Vs “Inform” users
- **Targets**
  - Content-makers, commentators, readers
  - Politicians, activists, dissidents, supporters
- **Easy to Switch Tactics**
  - Flooding, Threats, Distraction, Persuasion
  - Mix & Match

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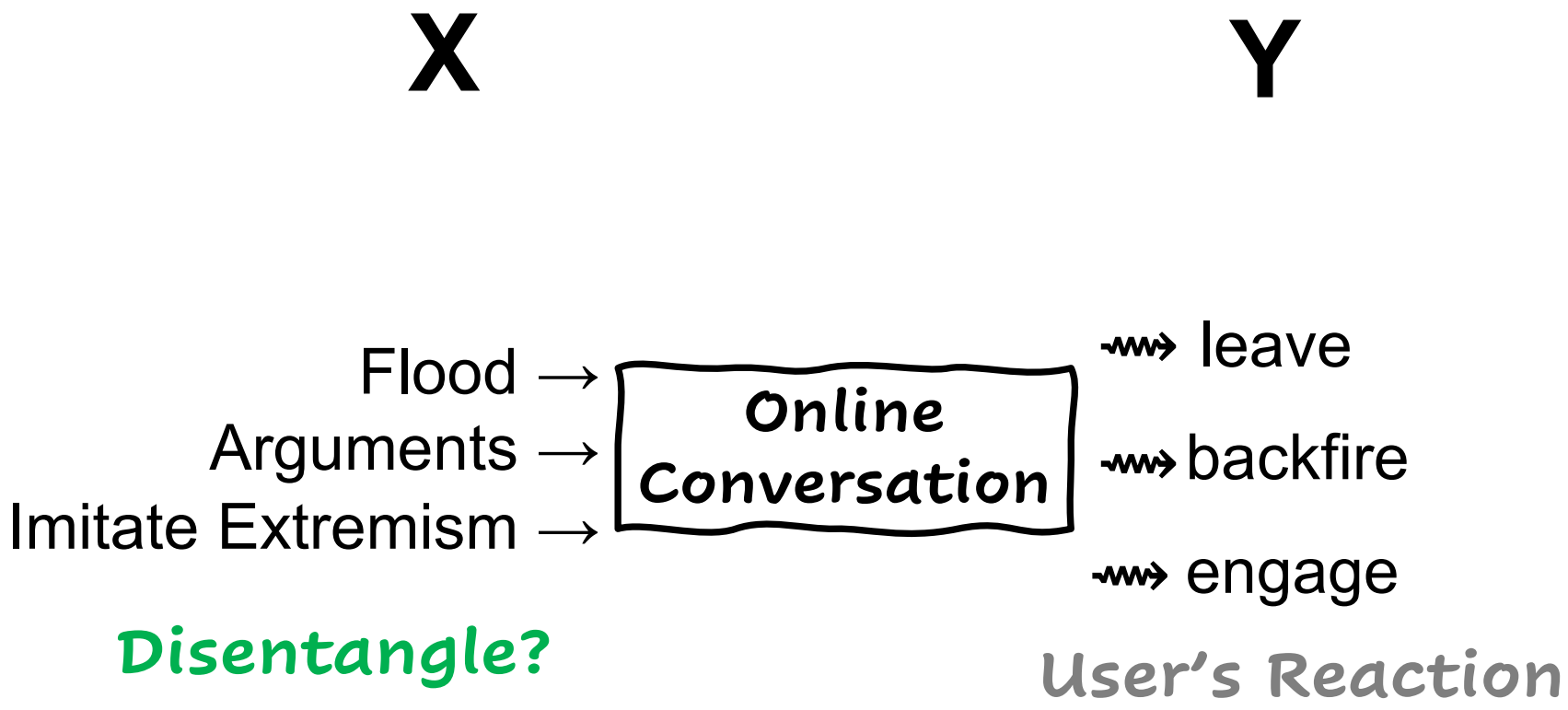
- **Masked Engagement**
  - Hide Troll Identity
  - Mimic Regular Users

# 2 Setting


X
Y
Flood →
Arguments →
Imitate Extremism →
Online Conversation
leave
backfire
engage
Disentangle?
User's Reaction

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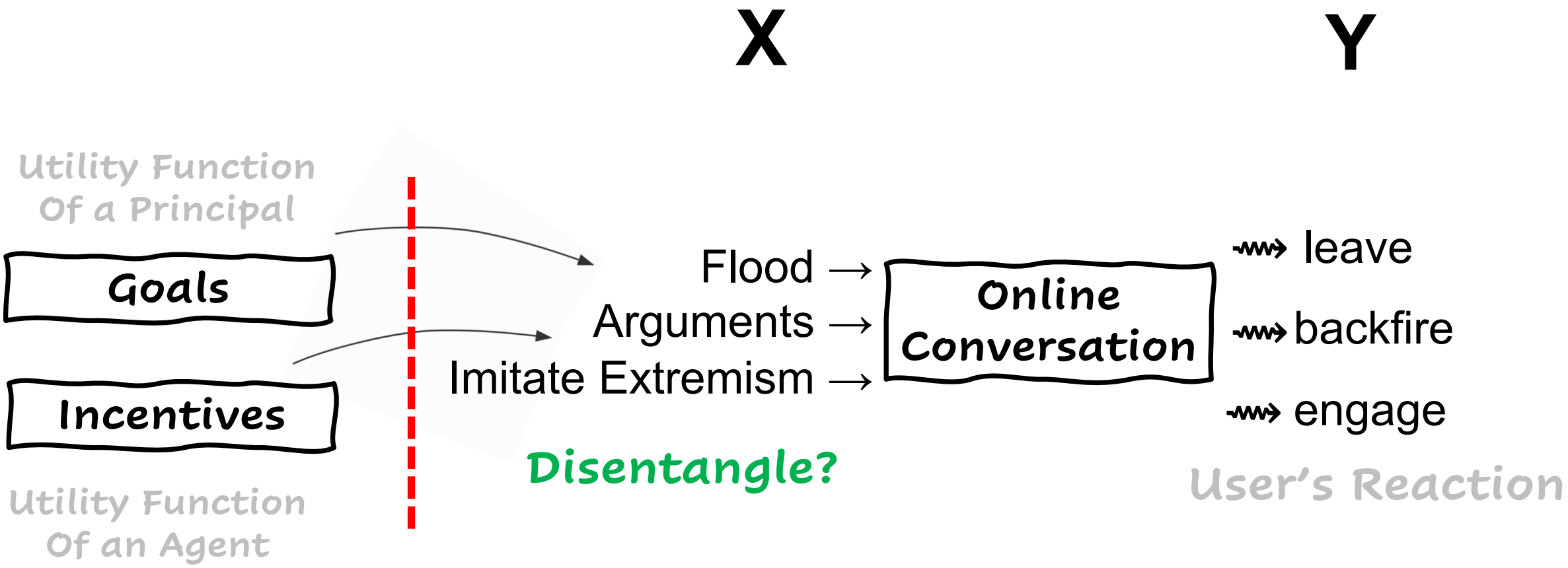
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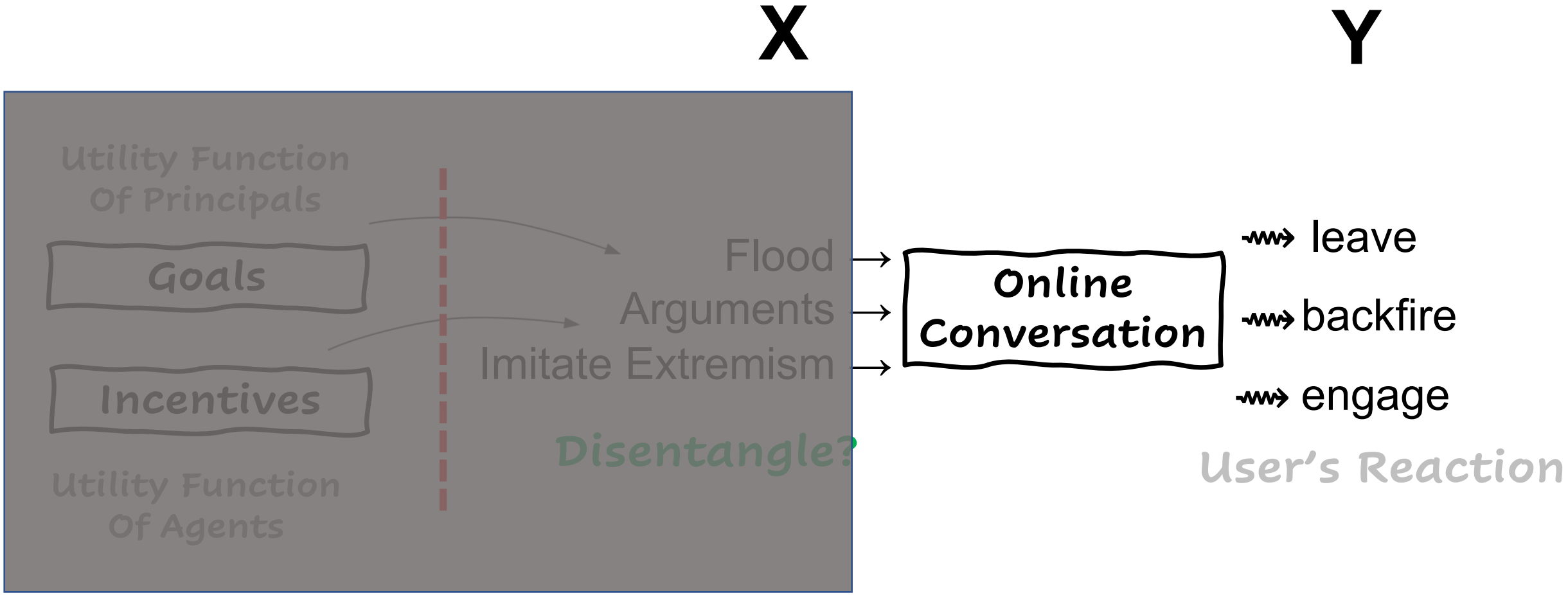



**Goal-1:** Estimate the overall observed effect of troll interventions
**Goal-2:** Estimate conditional effects

# Data

## ☐ Troll Factory Data Leak

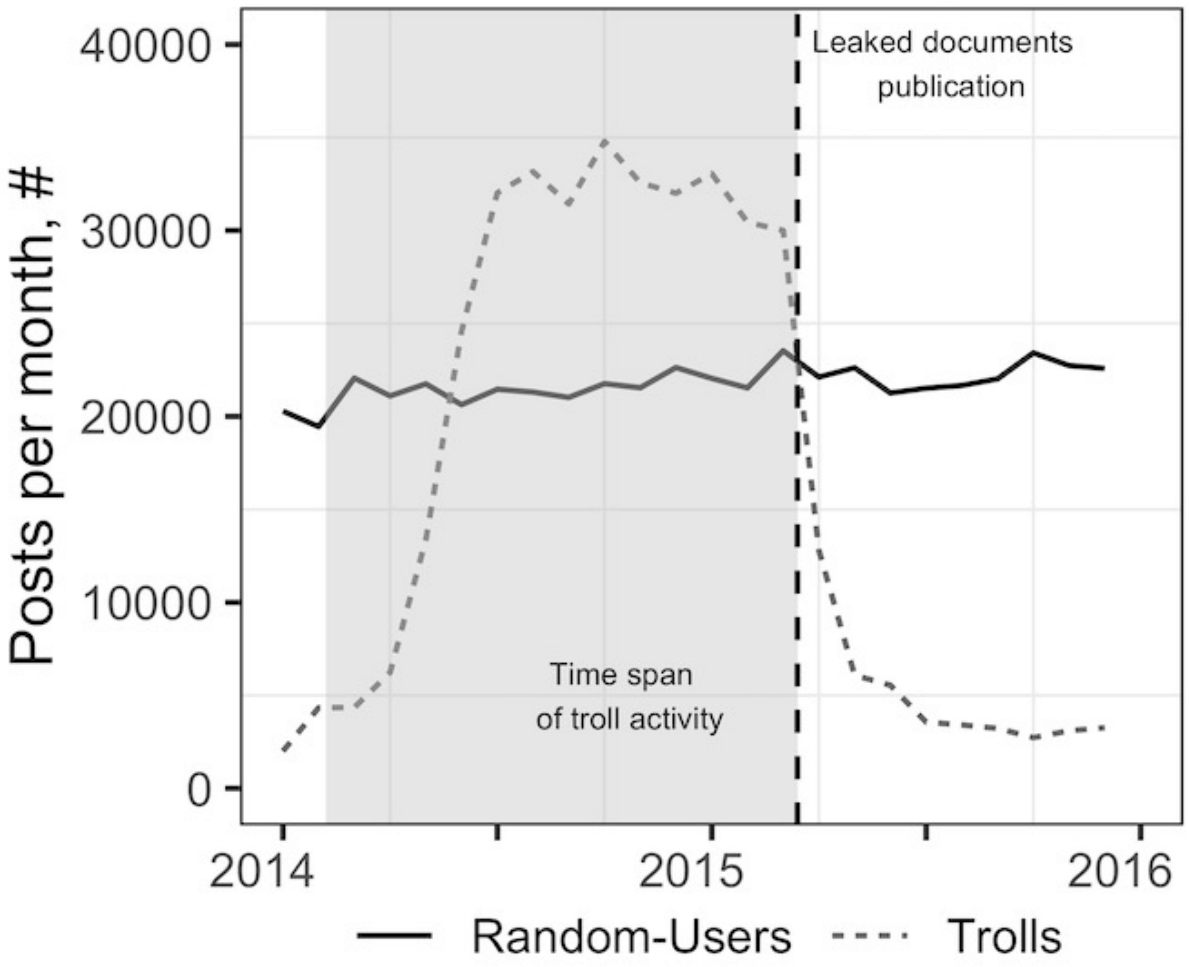
- 700 Leaked Accounts on *LJ*
- 2014-15 [Russia-Ukraine Conflict]
- 130k troll-infilitrated discussions

## ☐ Random *LJ* Users

- 4M Random Posts & Discussions

## ☐ Trolls' Target

- Russia's Domestic Audience [dissatisfied users]



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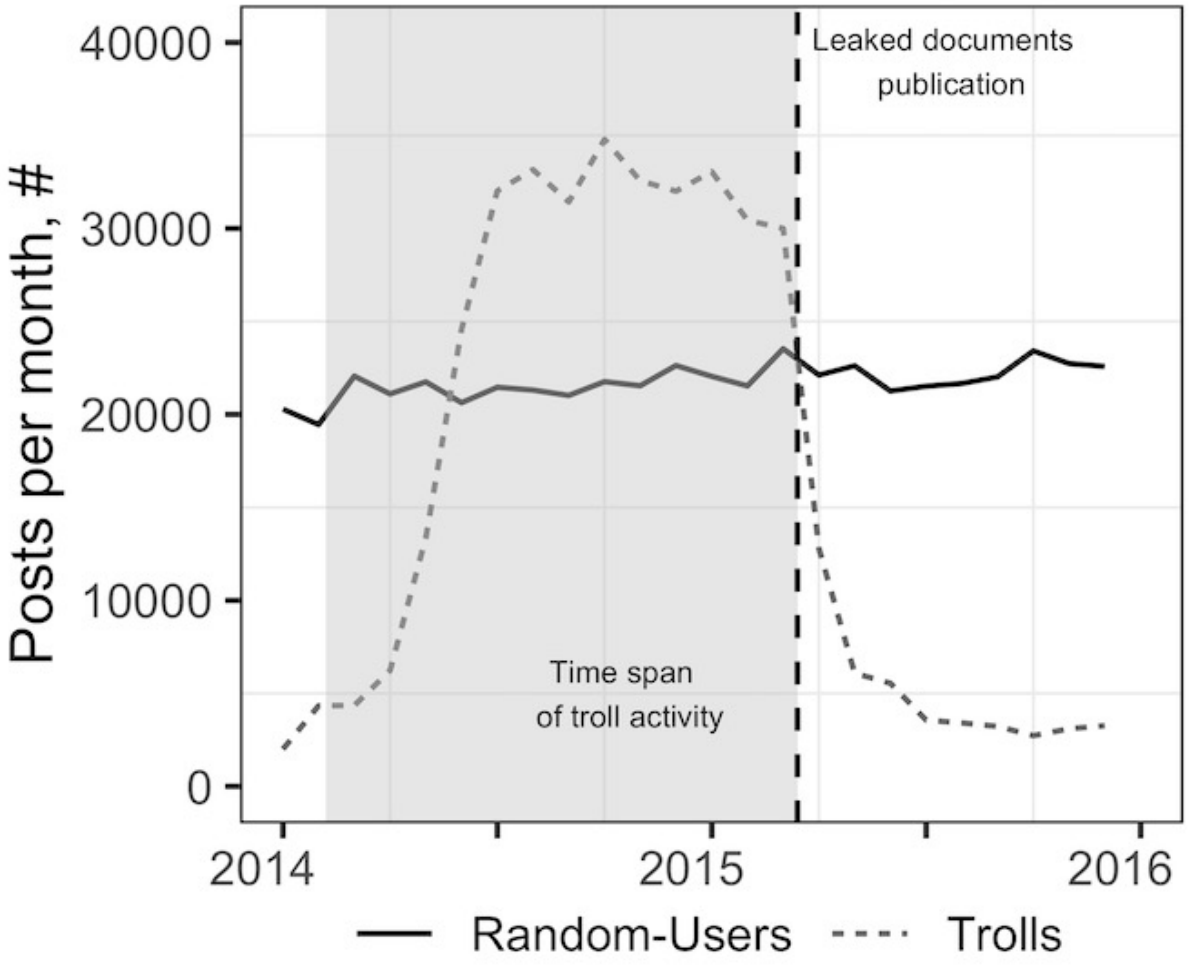



**Reason of Data Leak: Utility Function Of an Agent!**

# Research Design: Advantages of *LJ*

## **A1:** Very Simplified Commenting

**Decreased intensity of communication** [Translated from Russian]

A Activist A LIVEJOURNAL

**Putin annexed Crimea: What should we do?** Apr 10, 2014 21:01

Russian President annexed Crimea out of a long-standing desire to resuscitate the Soviet empire, and he may eventually go after the rest of Ukraine, as well as other countries in eastern Europe ( Read more... )

#Putin #Politics

LEAVE A COMMENT

Comments 142

**kinofriend** 20:49:23 UTC
His ambitions are too dangerous for Russia. What is next, **war**?

**nep-glad** 21:01:24 UTC
Agree. He'll get Russia into the new **war**

**daslight** 21:16:15 UTC
Probably he needs a **war** to distract the population

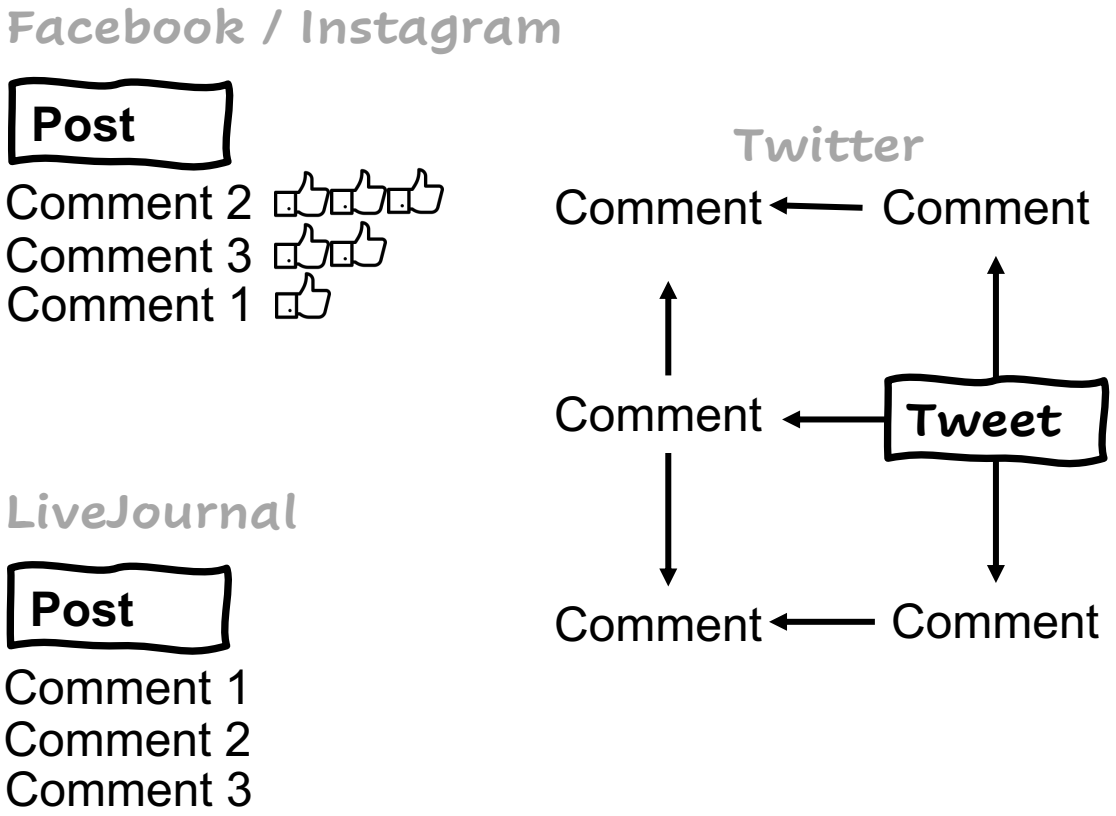
**deputat** 22:29:10 UTC
Looks like he sends **military** weapons to East Ukraine

**ilyapetrov** 22:30:10 UTC
You all are traitors!

**ilyapetrov** 22:31:11 UTC
Only Putin can save Russia

**yehu** 22:32:10 UTC
Probably he wants a civil **war** in East Ukraine

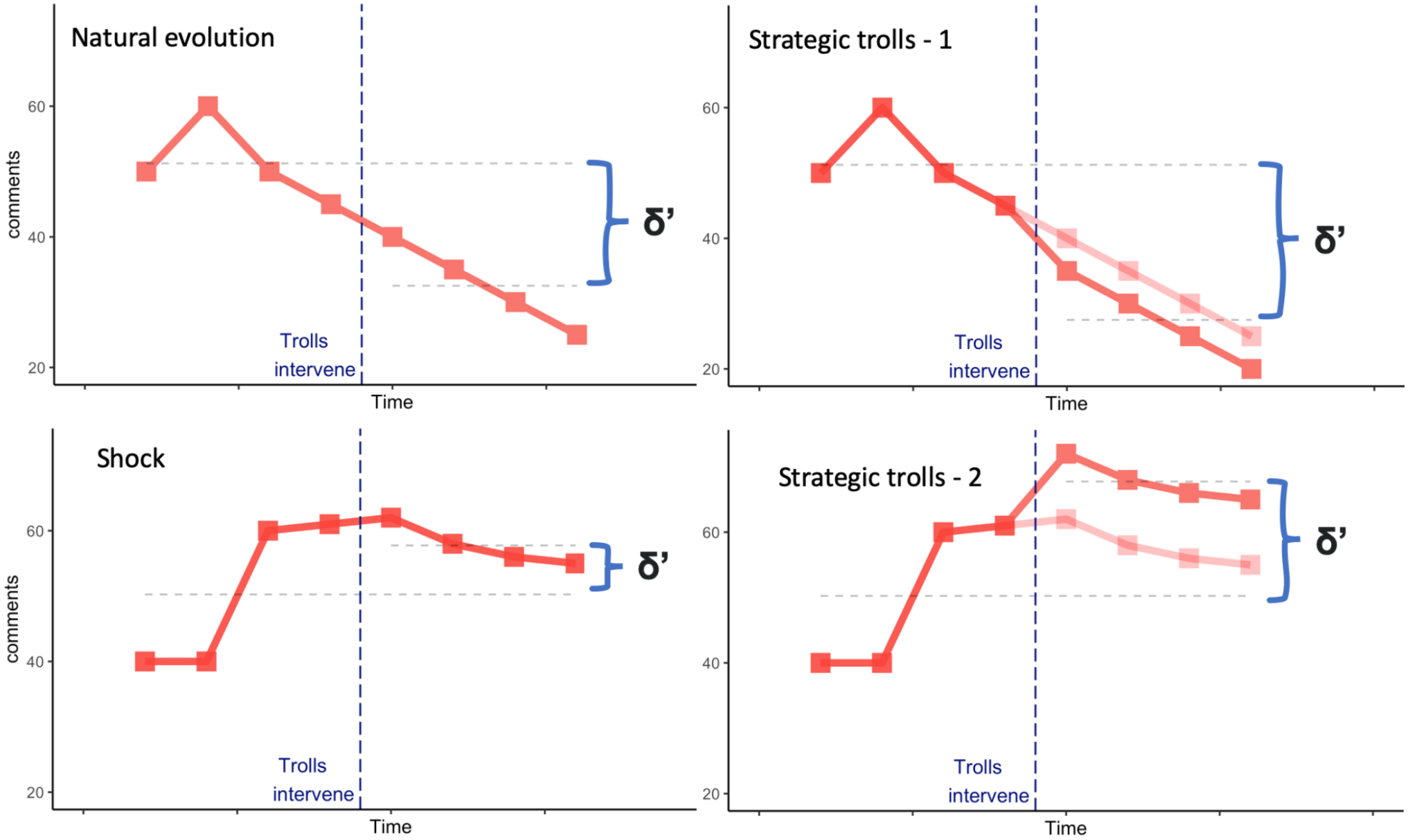
Discussion ends here




## **A2:** Very Simplified Search Engine [Post contents only]

## **A3:** "Read Random Post" feature [Sampling]

# Causal Inference Challenge



# Ideal Empirical Setting [desired]

## ☐ Observational Setting

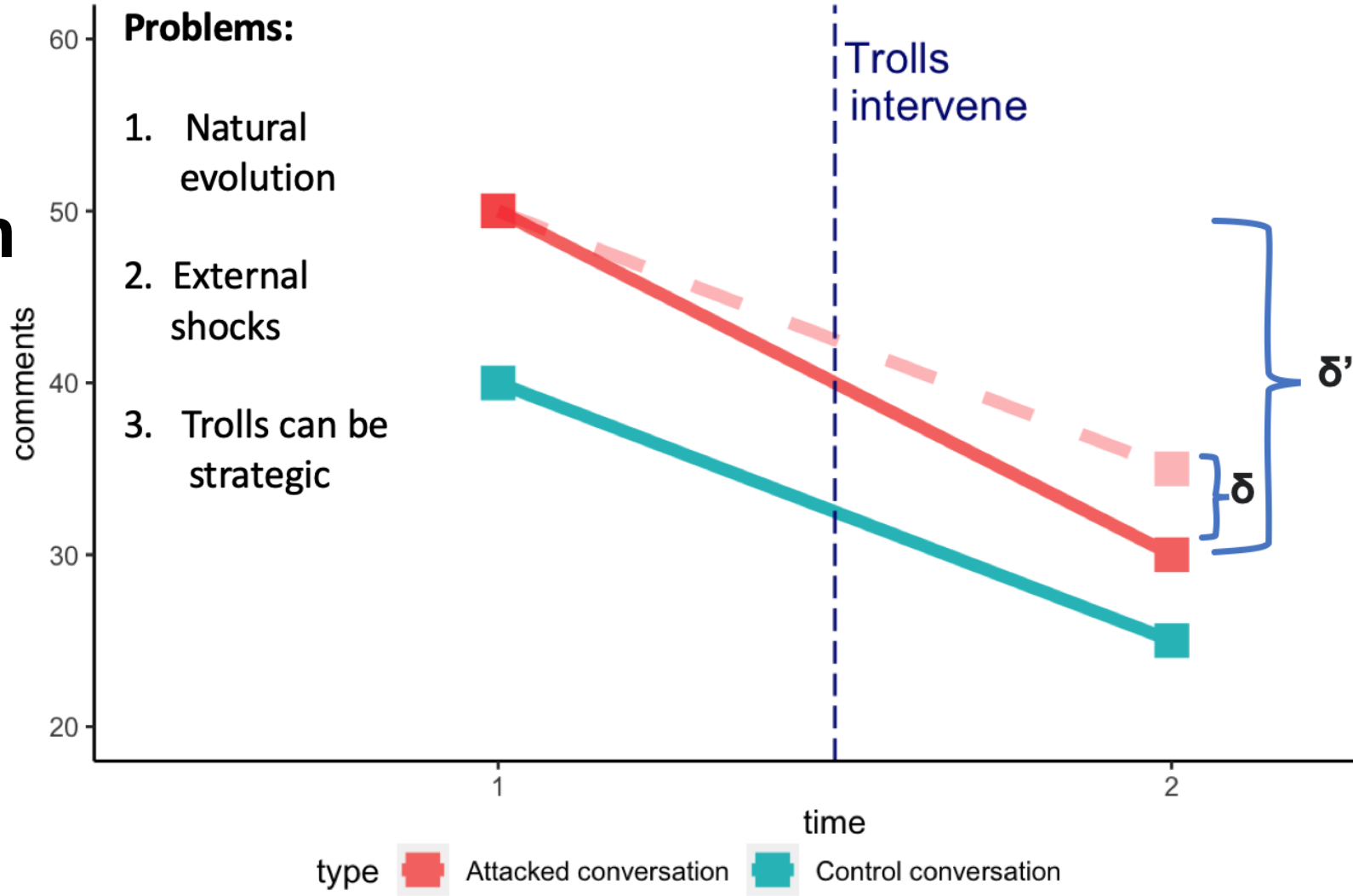
- DiD
- Matching

## ☐ Control Conversation

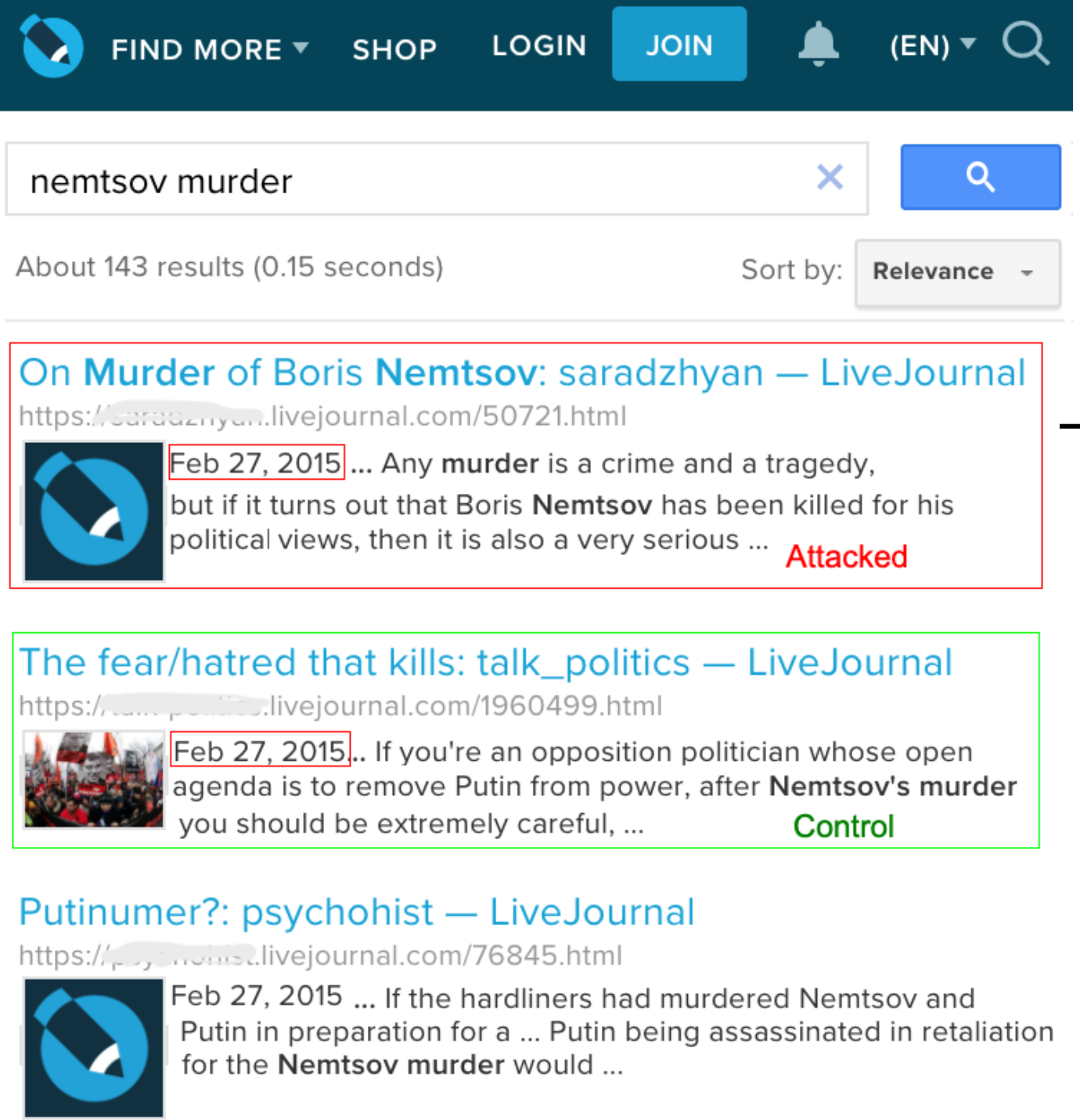
- Discuss the same topic
- High chance of attack
- Time overlap

## ☐ How?

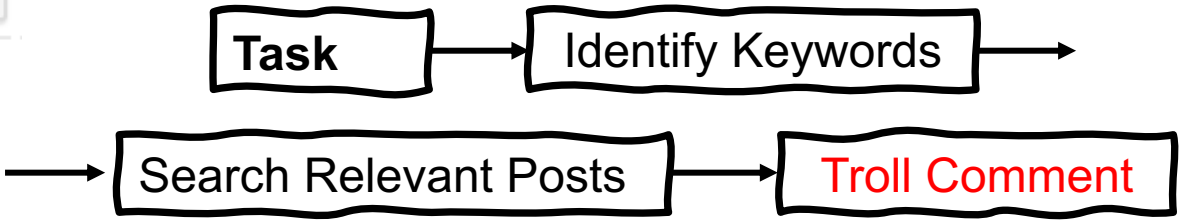
- Approximate DGP
- Simplicity of LJ Platform



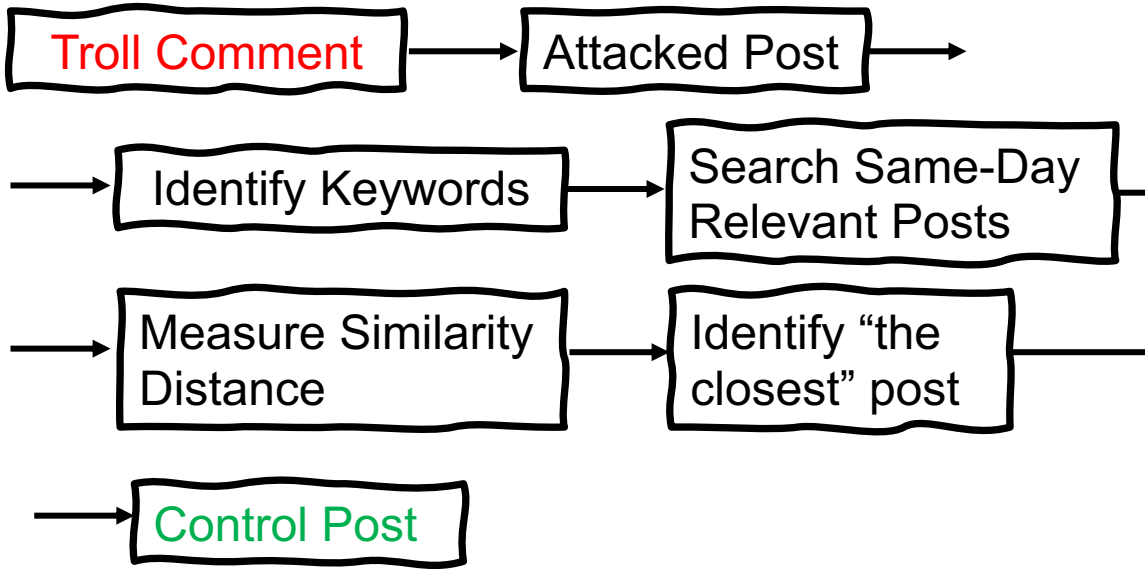
# Approximate Selection of Posts for Attack



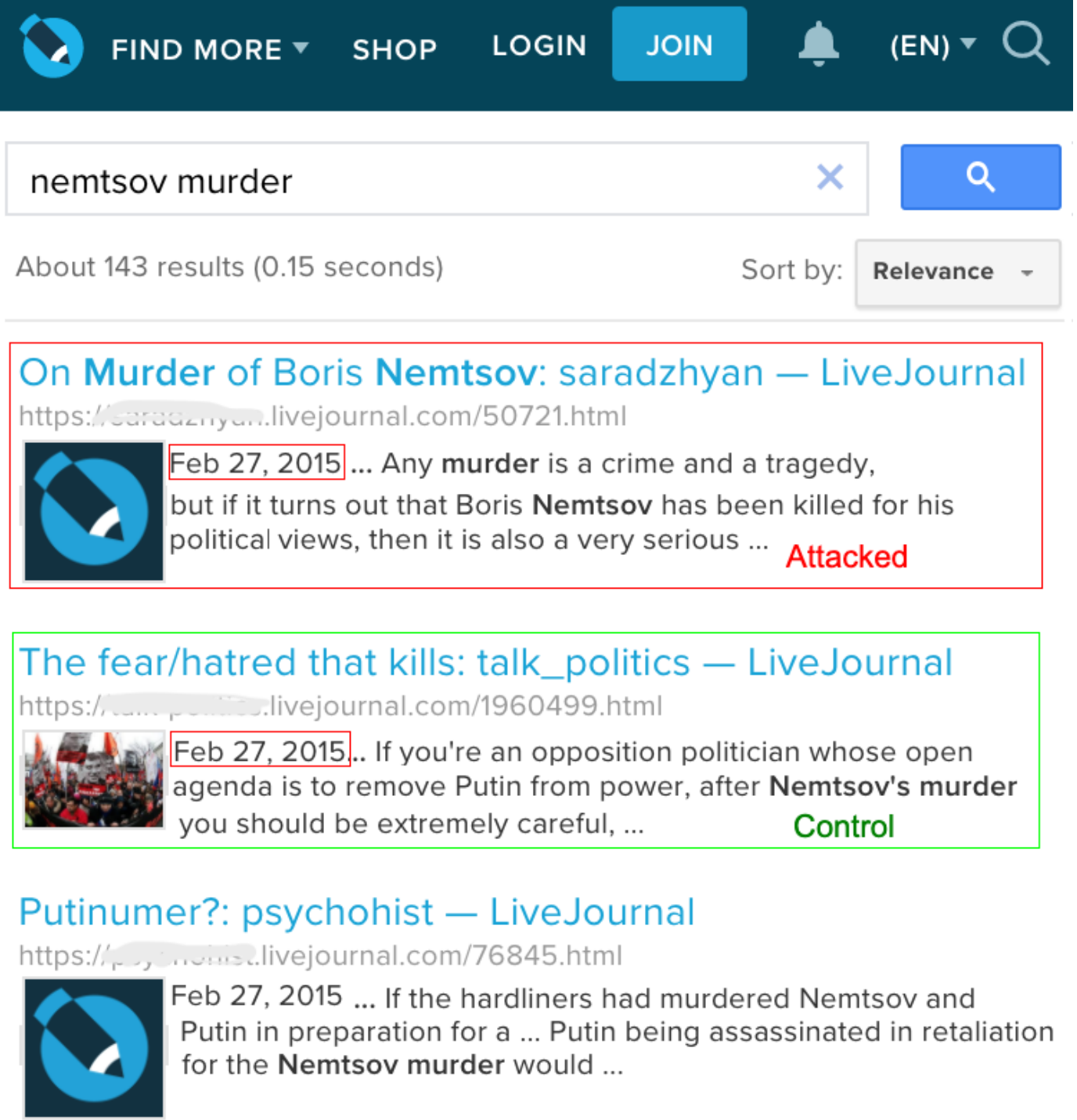

## Troll's Pipeline

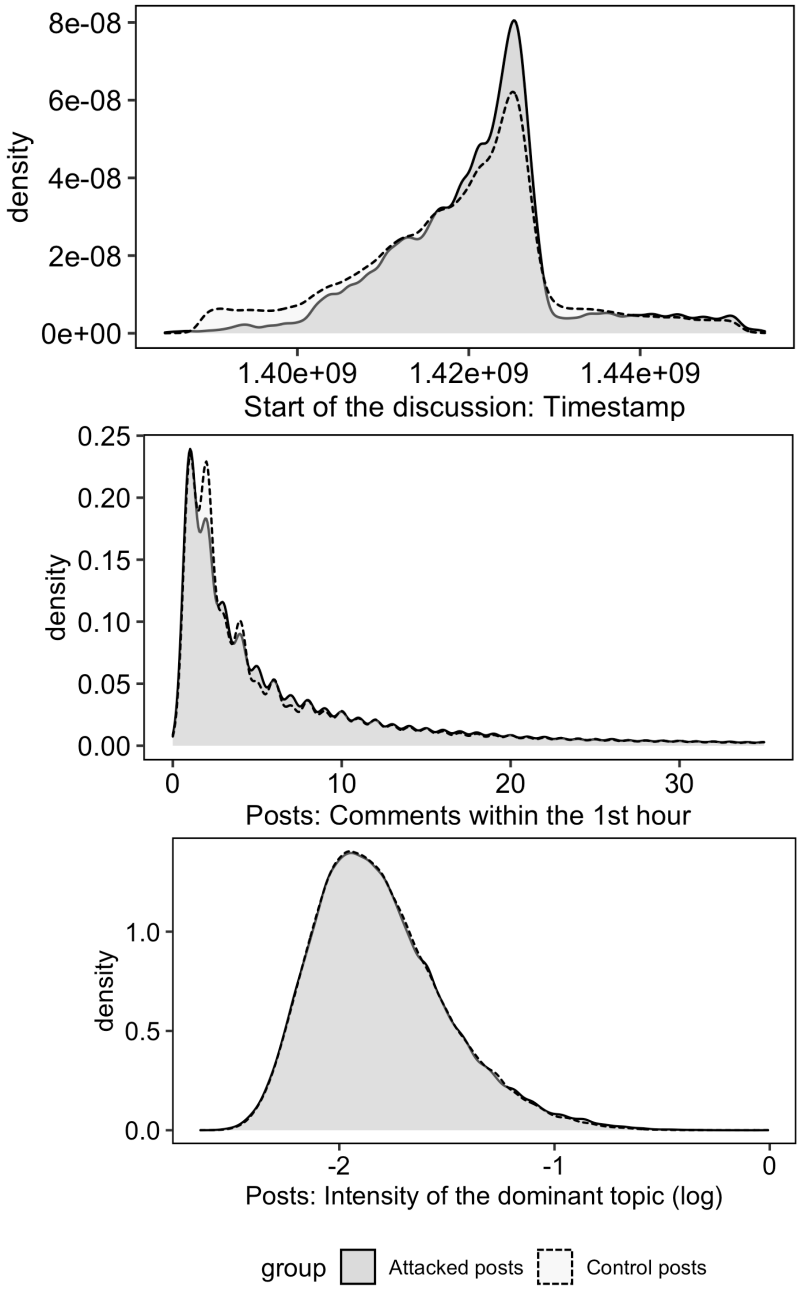



## Researcher's Pipeline



# Balance


FIND MORE
SHOP
LOGIN
JOIN
(EN)
nemtsov murder
About 143 results (0.15 seconds)
Sort by:
Relevance
On Murder of Boris Nemtsov: saradzhyan — LiveJournal
https:/ .livejournal.com/50721.html
Feb 27, 2015 ... Any murder is a crime and a tragedy, but if it turns out that Boris Nemtsov has been killed for his political views, then it is also a very serious ...
Attacked
The fear/hatred that kills: talk_politics — LiveJournal
https:/ .livejournal.com/1960499.html
Feb 27, 2015 .. If you're an opposition politician whose open agenda is to remove Putin from power, after Nemtsov's murder you should be extremely careful, ...
Control
Putinumer?: psychohist — LiveJournal
https:/ .livejournal.com/76845.html
Feb 27, 2015 ... If the hardliners had murdered Nemtsov and Putin in preparation for a ... Putin being assassinated in retaliation for the Nemtsov murder would ...


density
8e-08
6e-08
4e-08
2e-08
0e+00
1.40e+09
1.42e+09
1.44e+09
Start of the discussion: Timestamp
density
0.25
0.20
0.15
0.10
0.05
0.00
0
10
20
30
Posts: Comments within the 1st hour
density
1.0
0.5
0.0
-2
-1
0
Posts: Intensity of the dominant topic (log)
group
Attacked posts
Control posts

# Outcomes of Interest

## ☐ Structural Outcomes

- **Structure:** Comments & Threads
- **Intensity:** Users & Replies

[Coercion]

## ☐ Content Outcomes

- **Sensitive Topics**
  Political Opposition, Economy in Crisis, War in Ukraine
- **Agenda Promotion**
  NATO Aggression
  Putin's Foreign Policy Success

[Engagement]

| Initial post | | |
|---|---|---|
| User 1 | 1:12 | Comment 1 |
| User 2 | 1:14 | + Comment 2 |
| User 1 | 1:18 | Comment 3 |
| User 3 | 1:19 | Comment 4 |
| *Troll 1* | ~~1:25~~ | ~~Comment 5~~ |
| User 2 | 1:30 | Comment 6 |
| *Troll 2* | ~~1:37~~ | ~~Comment 7~~ |
| User 1 | 1:39 | Comment 8 |
| User 3 | 1:41 | + Comment 9 |
| User 2 | 1:43 | ++ Comment 10 |
| User 4 | 1:45 | ++++ Comment 11 |

| Matched control post | | |
|---|---|---|
| User A | 1:01 | Comment 1 |
| User B | 1:19 | + Comment 2 |
| User A | 1:21 | Comment 3 |
| User C | 1:30 | Comment 4 |
| User A | 1:32 | + Comment 5 |
| User B | 1:42 | Comment 6 |
| User C | 1:55 | + Comment 7 |

$$Y_{it} = \alpha + \beta Attacked_i + \gamma Period_t + \delta Attacked_i Period_t + \epsilon_{it}$$

# Results–1: Structural Outcomes

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
|---|---|---|---|---|---|---|---|---|
| | Comments | | Unique users | | Replies | | Threads | |
| Attacked discussions×Time | -26.01*** | -21.89*** | -6.57*** | -4.95*** | -20.15*** | -17.64*** | -12.90*** | -11.73*** |
| | (0.61) | (2.19) | (0.15) | (0.70) | (0.42) | (1.58) | (0.37) | (1.10) |
| Time: Post-Treatment | -5.71*** | -10.72*** | -3.95*** | -5.85*** | -0.67** | -3.77*** | -0.29 | -1.83*** |
| | (0.44) | (1.50) | (0.11) | (0.47) | (0.30) | (0.98) | (0.27) | (0.64) |
| Attacked discussions | -2.53*** | -5.262*** | -4.32*** | -4.37*** | 2.50*** | -1.17 | 4.51*** | -0.13 |
| | (0.411) | (1.47) | (0.10) | (0.54) | (0.28) | (1.00) | (0.25) | (0.75) |
| Constant | 39.57*** | | 17.45*** | | 21.91*** | | 10.68*** | |
| | (0.29) | | (0.07) | | (0.20) | | (0.18) | |
| Author Fixed Effects | | ✓ | | ✓ | | ✓ | | ✓ |
| Clustered Standard Errors | | ✓ | | ✓ | | ✓ | | ✓ |
| Observations | 510,935 | 510,935 | 510,395 | 510,935 | 510,395 | 510,935 | 510,395 | 510,935 |
| $R^2$ | 0.02 | 0.15 | 0.04 | 0.28 | 0.01 | 0.14 | 0.01 | 0.08 |

*Note:* In models (2), (4), (6), (8) standard errors are clustered by authors of the initial posts. ***$p < .01$, **$p < .05$, *$p < .1$.

# Results–2: Content Outcomes

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) |
|---|---|---|---|---|---|---|---|---|---|---|
| | Politically sensitive topics | | | | | | | | Pro-government topics | |
| | Ukraine | | History | | Opposition | | National economy | | Putin's politics | |
| Attacked discussions×Time | -0.12*** | -0.11*** | -0.21*** | -0.18*** | -0.10*** | -0.09*** | -0.10*** | -0.09*** | 0.02*** | 0.01 |
| | (0.01) | (0.01) | (0.00) | (0.01) | (0.00) | (0.01) | (0.00) | (0.01) | (0.00) | (0.01) |
| Time: Post-Treatment | -0.01** | -0.02*** | -0.06*** | -0.09** | -0.03*** | -0.04*** | -0.01** | -0.02*** | 0.04*** | 0.05*** |
| | (0.00) | (0.00) | (0.00) | (0.01) | (0.00) | (0.01) | (0.00) | (0.00) | (0.00) | (0.01) |
| Attacked discussions | 0.08*** | 0.09*** | 0.12*** | 0.2*** | 0.01* | -0.03** | 0.07*** | 0.05* | -0.02*** | 0.00 |
| | (0.00) | (0.01) | (0.00) | (0.01) | (0.00) | (0.01) | (0.00) | (0.02) | (0.00) | (0.01) |
| Constant | -3.62*** | | -2.72*** | | -3.01*** | | -3.347*** | | -3.40*** | |
| | (0.00) | | (0.00) | | (0.00) | | (0.00) | | (0.00) | |
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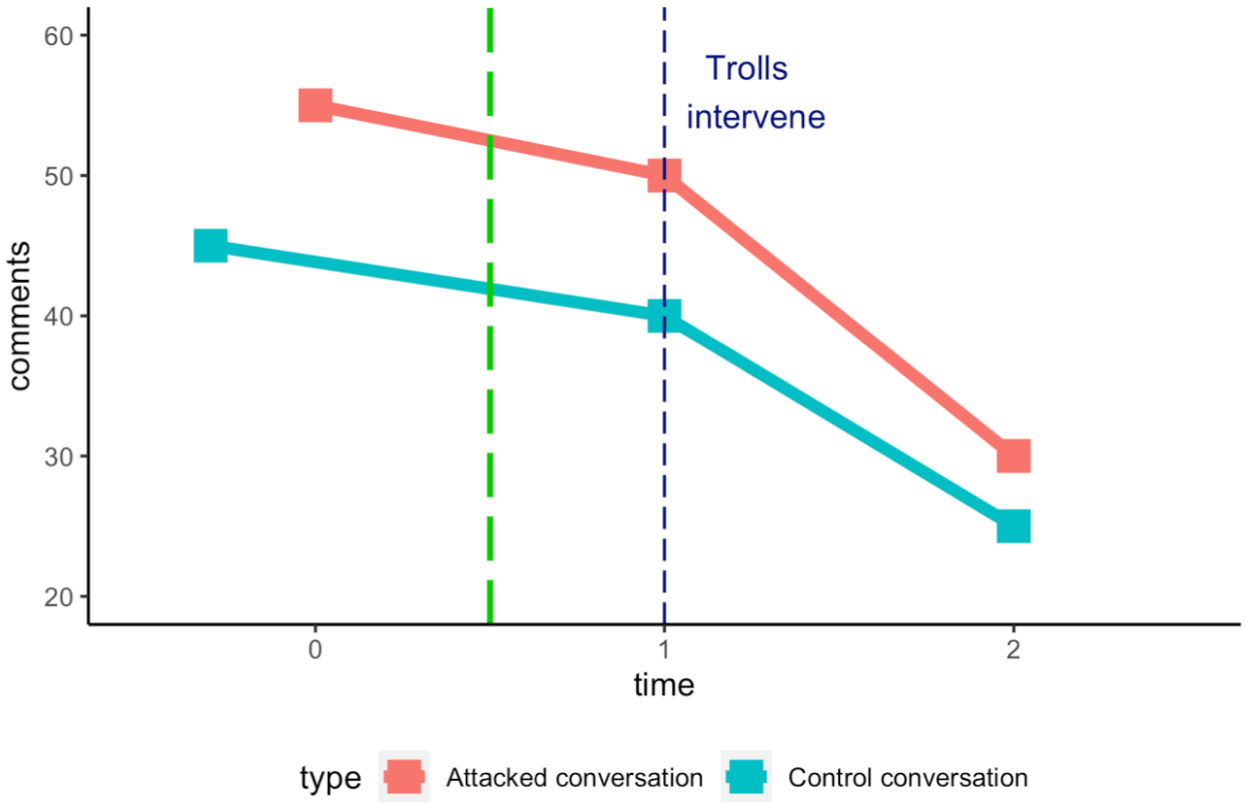
# Threats to Validity

- ☐ **Underlying parallel trends**
- ☐ Effect of a new user
- ☐ Deleted comments
- ☐ Incomplete List of Trolls



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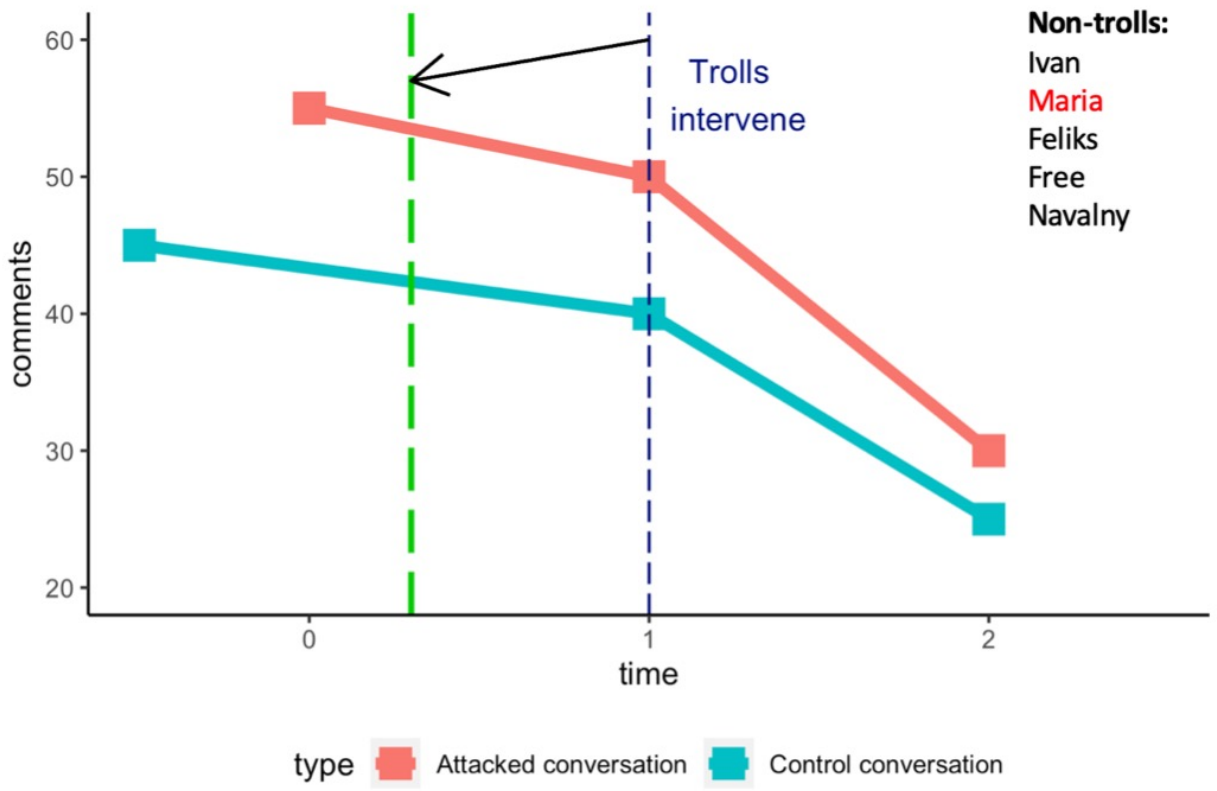

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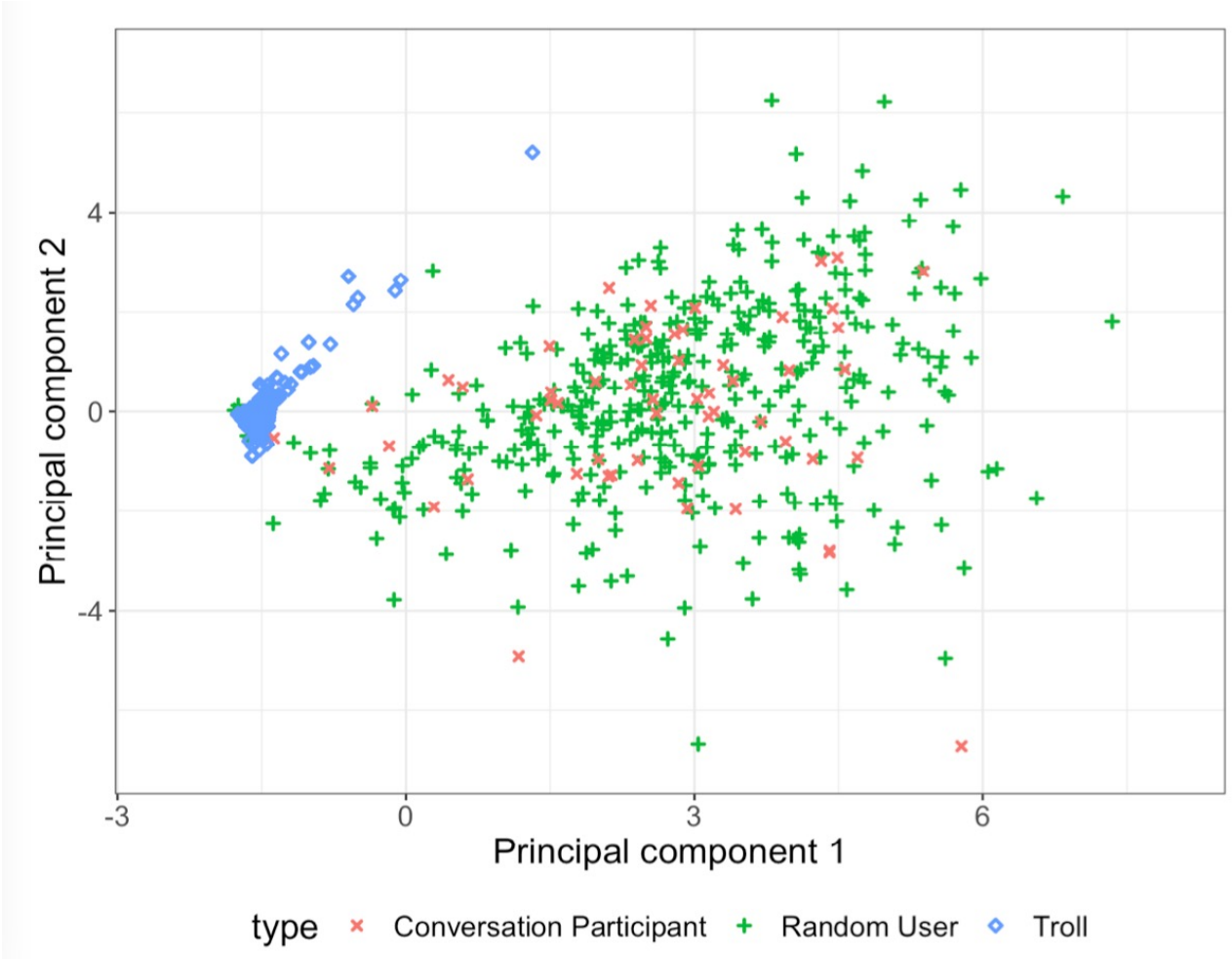
- ☐ Underlying parallel trends
- ☐ Effect of a new user
- ☐ Deleted comments
- ☐ **Incomplete List of Trolls**

| Monday | Tuesday | Wednesday | Thursday | Friday | Saturday | Sunday |
|---|---|---|---|---|---|---|
| Group 1 | Group 3 | Group 1 | Group 3 | Group 1 | Group 3 | Group 1 |
| Group 2 | Group 4 | Group 2 | Group 4 | Group 2 | Group 4 | Group 2 |

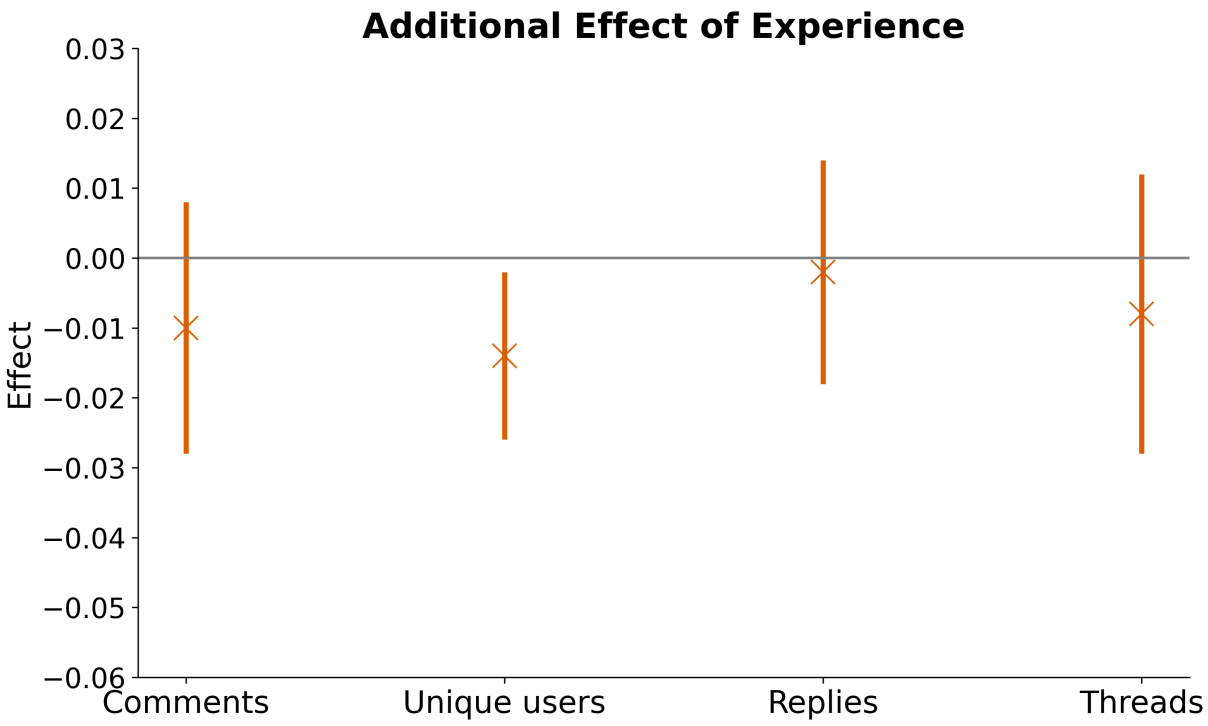


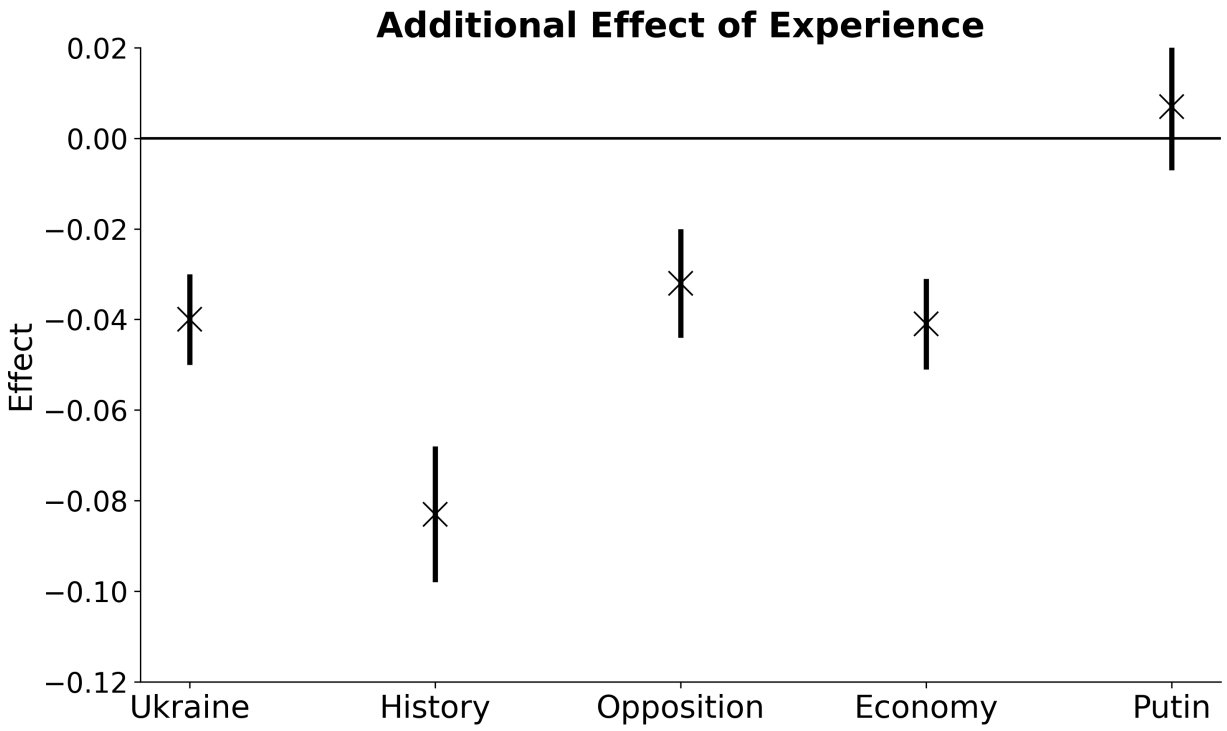
# “Narrow” Conclusions

- **Identified Effects of Troll Interventions**
  - Trolls prevent political communication
  - Effect is amplified
  - No evidence on promotion of pro-government agenda
- **Approximation of DGP**
  - Specifics of the Cyberspace
  - Simplicity of LJ platform engine

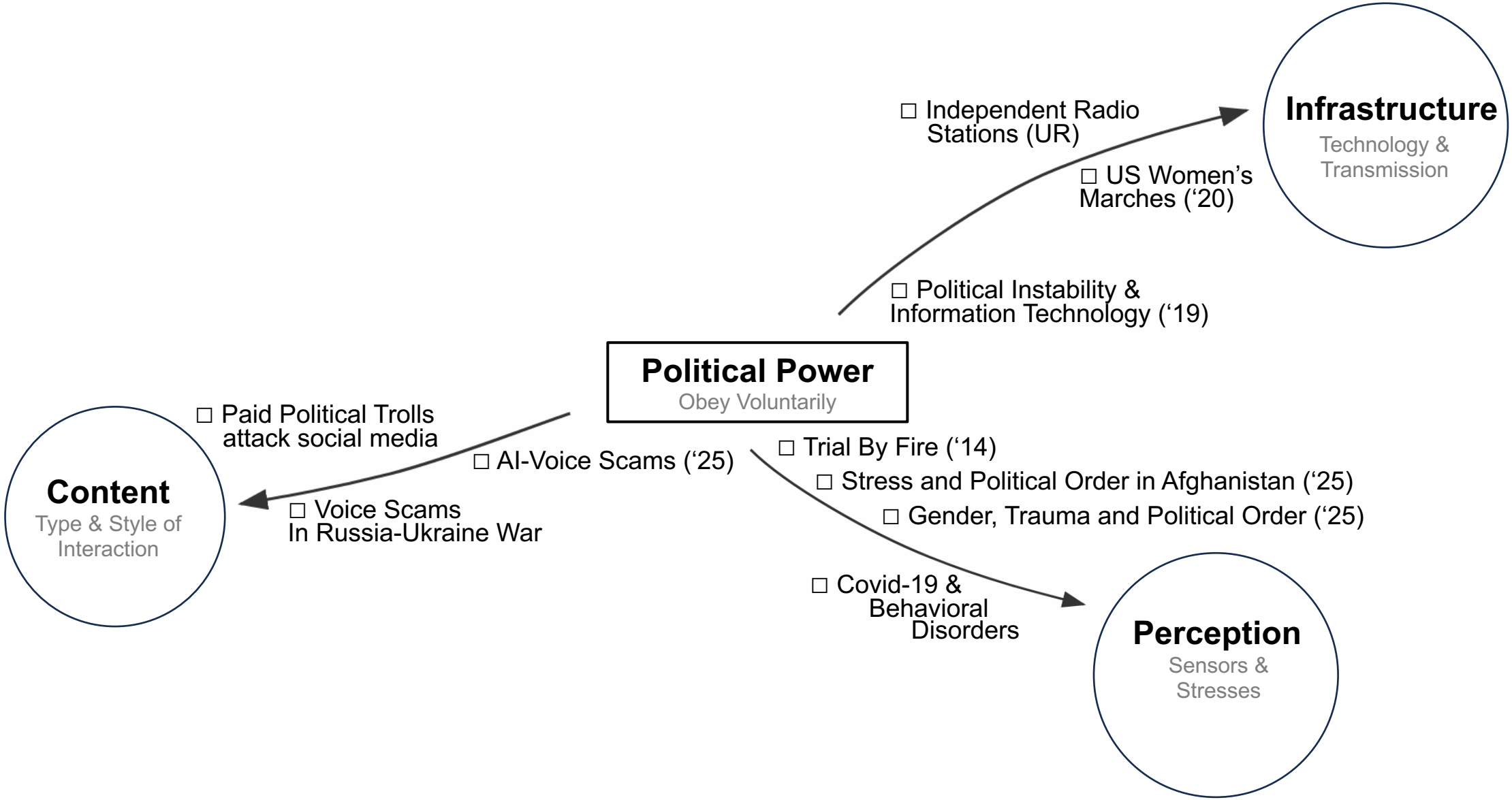
# Conditional Effect

## Personal Experience & Opposition Views



# Power & Information



# “Broad” Speculations

## ☐ “Troll Effect” ⇝ Political Power

- A ⇝ X | B
- No coercion observed
- Legitimacy is required

## ☐ Types of Legitimacy Weber 1919

- Traditional
- Charismatic legitimacy
- Legal-rational legitimacy